

AI-Powered Interview Preparation and Resume Coach

Mrs.S Punitha

Associate Professor
Computer Science and
Engineering,
Rajalakshmi
Engineering College
Chennai, India

punitha.s@rajalakshmi.edu.in

Kaveeshaa S

Student
Computer Science and
Engineering
Rajalakshmi
Engineering College
Chennai, India

230701146@rajalakshmi.edu.in

Krithika PV

Student
Computer Science and
Engineering
Rajalakshmi
Engineering College
Chennai, India

230701157@rajalakshmi.edu.in

Abstract - Moving from school to work jobs needs students to learn skills. These skills include writing a resume knowing facts, coding, solving problems talking well and getting ready for interviews. But usual ways to prepare for placement give personal help and do not give a single place to test these skills. The proposed system has parts. These parts include resume analysis, Applicant Tracking System scoring, job description matching, mock interviews, technical interviews, communication evaluation, coding practice, aptitude tests, skill recommendations, personalized learning paths and performance tracking. The system reads PDFs and DOCX files pulls out the text and checks resume details such as keywords, sections, work experience, projects, terms and contact information. The ATS part uses a six-step score. Job descriptions are reviewed to find skills and keywords and then compared with the candidate's resume. The platform works with two AI modes. The default mode uses a rule-based engine and an optional mode can use a large language model. User data is saved in Supabase PostgreSQL. Is used to give personalized suggestions and learning plans. Security features such as JWT authentication, password hashing, rate limiting, HTTP security headers, file checks and ownership verification are built in. The proposed system gives students a data-driven place to boost their readiness, for placement.

Keywords— Artificial Intelligence, Resume Analysis, ATS Interview Preparation, Natural Language Processing, Job Matching, Coding Practice, Aptitude, Personalized Learning

I. INTRODUCTION

In today's job market students need more than just academic knowledge to do well in technical placement

processes. Employers look for a mix of skills coding ability, aptitude, communication skills, interview performance and a strong resume.. Most students prepare for each of these areas separately. They use platforms for coding practice resume writing, mock interviews and skill tests. This scattered approach makes it hard for students to see where they stand overall.

Resume screening is one of the steps in hiring. Companies often use Applicant Tracking Systems (ATS) to scan resumes. These systems check for keywords, relevant experience, qualifications and how well the resume fits the job description. Many students don't know how ATS works. Because of this their resumes may not pass through these automated filters even if they are qualified. Similarly popular interview prep tools usually offer sample questions and practice sessions.. They rarely give real feedback or help track progress over time. Students can't see how their performance changes or what specific skills they need to improve.

To solve this problem we are building an AI-Powered Interview Preparation and Resume Coach. It's a web-based platform that brings together all key placement preparation activities in one place. The platform helps with resume analysis, ATS scoring matching resumes to job descriptions mock interviews, technical interviews, communication evaluation, coding practice, aptitude tests, skill recommendations and personalized learning paths. The system analyzes uploaded resumes by pulling out details detecting important keywords checking how complete each section is and offering suggestions for improvement. It also gives an ATS score based on six main factors: keyword matching, experience and projects section completeness, structure, terminology and contact information.

Beyond that the platform keeps track of user activity and performance. It uses this data to find gaps in skills. Suggest improvements tailored to individual needs. A central dashboard shows users their placement-readiness score, how they're doing in each skill area their learning journey far and past tasks completed. The platform is built using a full-stack architecture. It uses React for the frontend, Node.js and Express for the backend and Supabase PostgreSQL for storing data. It also includes a mode AI system. One part uses a built-in engine for structured evaluations and the other allows integration with OpenAI-compatible large language models when needed.

This unified system aims to give students a more personal and centralized way to prepare for placements. Of juggling many tools students can focus on one intelligent platform that guides them step by step toward better readiness, for technical jobs.

II. LITERATURE REVIEW

The paper (Kadam et al. 2022) Presents an NLP-Based Resume Screening and Job Recruitment Portal. This portal is designed to automate the extraction and analysis of data from resumes. It uses methods such as expressions, pattern searching, tokenization and NLP libraries to spot relevant details in resumes. The work shows how unstructured resume files can turn into data useful for hiring. The portal helps cut the work needed to sort many resumes.. This portal mainly deals with resume screening and extraction. It does not offer an environment that covers interview preparation, coding practice, aptitude tests, communication skills or personal learning.

The paper (Borude 2025) proposes an AI-based resume screening system that uses Natural Language Processing and Machine Learning. The system parses resumes pulls out information and compares candidates with job descriptions. It applies comparison with word embeddings along with checks for relevance, experience and skills to rank candidates. Unlike keyword filters this system tries to understand the context, between a candidate's profile and the job needs. It improves the stages of hiring by automating candidate selection from a recruiter's point of view. However it does not supply a

student-focused platform that tracks performance supports ongoing placement prep or offers personalized learning.

The paper (Chong, Ng and Fu 2025) talks about an AI system for hiring that brings together checking resumes and a chatbot for job recruitment. The part that checks resumes uses ways to read documents and the power of language models to pull out and make sense of the information in the resumes. This information is then checked against what the job needs to see how well the person fits. The system also has a chatbot that can answer questions and change voice into text to make the hiring process more friendly. Combining the ability to read documents using AI to check resumes and having a chatbot makes the hiring process better. However the system is mainly for helping with hiring. Does not look at how well a candidate can code what their skills are, how they do in technical interviews or how they talk over time.

The paper (Kushwaha et al. 2026) Looks at AI systems for checking resumes and matching people to jobs. It uses tools like natural language processing, deep learning, transformer models and fuzzy logic. The paper talks about the problems with systems that only look for keywords and looks at new ways to understand resumes and job ads better. The research also points out that being able to explain decisions and stop bias is important in these systems. These methods can make the hiring process more accurate and easier to use. However the complicated ways, with transformer models and mixed methods can need computer power and the research mainly focuses on hiring and picking people instead of helping students get better at finding jobs.

I read the paper by Zhu et al. 2025 Which looks at how Large Language Models can help job recommendation by completing resumes. The authors focus on personalized job recommendation. Check how LLM-based resume completion can make a user's profile line up better with the right jobs. The method shows that mixing resume details with recommendation can make job searching better. Still the system only looks at job recommendation. Does not cover interview skills, coding ability, aptitude, communication or overall readiness for placement.

I also read Szandała 2025, which looks at Large Language Models such as ChatGPT, Google Gemini and Mistral used to judge IT candidates against experts. The authors ask if

AI can measure competencies during hiring and compare AI scores to human scores. They find that AI evaluation is much faster than evaluation but they also spot differences, in how consistent AI and human evaluations are. This shows that LLMs could help with technical candidate evaluation. Yet the paper warns that AI assessment must be used carefully because consistency and reliability are still concerns.

The paper by Yazdani, Mahajan and Ansari in 2025 introduces Zara, a system that uses a language model, specifically GPT-4o to create personalized practice interviews. Zara conducts assessments and offers structured feedback to candidates. Zara also employs Retrieval-Augmented Generation to answer candidate questions. The study shows that large language models can provide personalized interview preparation without the need for a human interviewer during each practice session. However Zara focuses on interview preparation and feedback and Zara does not combine resume analysis, ATS scoring, coding practice, aptitude assessment or personalized learning paths into one placement platform.

The paper by Yuksel and colleagues in 2026 proposes an AI framework that uses large language models for candidate assessment. This framework gathers information from job descriptions, CVs, interview transcripts and HR feedback to produce candidate evaluation reports. Role-specific rubrics provide assessments moving beyond simple keyword-based ATS scoring. The framework enhances interpretability by delivering criteria-based evaluation reports and candidate comparisons. However this framework is designed for organizational recruitment and relies on a complex AI architecture. The framework does not address the learning and preparation needs of students preparing for campus placements.

The paper by Chuang and colleagues in 2021 studies bias mitigation in machine-learning based language classification using rationalization. The study shows how biases in training data can affect AI predictions and offers a method to separate relationships from spurious correlations. Although the research is not focused on recruitment the findings are relevant to candidate evaluation because automated recruitment systems can also be affected by data biases. The method improves fairness. Adds complexity and does not directly address resume analysis, interview preparation or personalized student learning.

Recent research titled "Towards Bias-Free Recruitment" in 2026 examines bias in LLM-based resume screening. The research highlights that while LLM systems can increase recruitment efficiency they may also reproduce biases from training data. The proposed adversarial contrastive tuning approach aims to reduce bias while maintaining the performance of the resume screening model. This work emphasizes fairness, transparency and responsible AI in automated recruitment. However the research focuses on bias mitigation in resume screening and does not provide a complete placement preparation environment.

From the literature above it can be seen that existing systems explore aspects of AI-assisted recruitment such, as resume parsing, resume screening, job matching, job recommendation, automated interviews and LLM-based candidate assessment.. Most approaches focus on a single recruitment stage or are designed mainly from the recruiters perspective. There is integration of resume improvement, ATS analysis, job matching, interview preparation, communication assessment, coding practice, aptitude testing and personalized learning within one student-oriented platform.

The proposed AI-Powered Interview Preparation & Resume Coach fills this gap by combining functions into a single placement preparation platform. The system extracts resume text analyzes keywords scores ATS compatibility, matches job descriptions evaluates interviews, assesses technical interviews, evaluates communication, provides coding practice and administers aptitude tests. Additionally the system records performance data. Uses actual metrics to spot skill gaps and create personalized learning paths.

The paper by Kadam et al. 2022 Describes an NLP-based resume screening system. This system pulls out information from candidate resumes and compares it with job requirements. It relies on natural language processing methods to automate the resume evaluation process, which helps cut down on effort.. The system is mostly limited to resume screening and recruitment. It does not offer support for interview preparation, coding assessments, aptitude tests, communication skills evaluation or personalized learning suggestions, for students.

III. METHODOLOGY

The proposed system is built as a full-stack web

application that contains connected modules designed to aid placement preparation. The system contains resume analysis, ATS scoring, job description matching, mock interview, technical interview, communication evaluation, coding practice, aptitude assessment, skill recommendation and personalized learning paths. All of these modules share a backend and database so the system can continuously record and analyze each user's performance. The system is created to provide students with one place to practice every stage of the placement process.

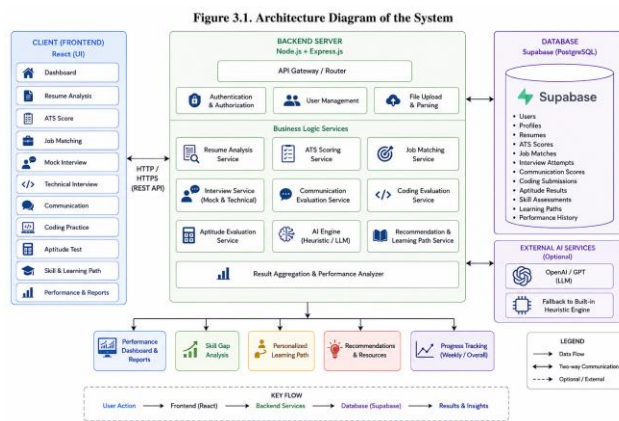


Figure 3.1. Architecture Diagram of the system

The front end is built using React, which creates the user interface that people use to access the parts of the application. The back end is built using Node.js and Express. Handles tasks like logging in processing data looking at resumes evaluating interviews and talking to the database. Supabase PostgreSQL is the database used to keep user details resume data, interview results, coding tests, aptitude tests, communication scores, skill checks and learning plans. The front end connects to the end using REST APIs and Axios. The first big part of the system is the Resume Analysis Module. This part takes resumes that users upload in PDF or DOCX format. The document is looked at to find text. That text is checked to find important details like name, email, phone number, education, work experience, projects, certificates, skills and languages. The system also looks at the words used to see which ones are important which are missing and which can be better. Based on what's found the system gives advice to help the user make their resume better, in quality and how it is put together.

RESUME AND ATS MODULE

The resume and ATS module also contains an ATS scoring part that checks a resume for the chosen job role. The scoring uses six parts: keyword matching, experience and projects section completeness, structure, terminology and contact information. The total score is out of one hundred. Each part shows a clear explanation of the result. The system also lists which keywords were found and which were missing so you can see where improvement is needed.

JOB MATCHING MODULE

The job matching module compares your resume with a job description. It pulls out skills and key words from that description. Check them against your resume. The module then gives a match score, lists matching skills, missing skills, matching keywords and offers suggestions. It also flags requirements such as experience, education and certifications helping you see how suitable your resume is for that job.

INTERVIEW MODULE

The Interview Module gives users a setting to practice both technical and behavioral interviews. Users can choose a job role, interview type, target technologies and difficulty level before users start an interview session. The system creates questions based on the chosen setup. Lets users submit answers by typing or speaking. This lets users answer questions in a clear interview-style way. Every response that a user submits is recorded and checked by the system. The system looks at things, such as, how long the answer is, how relevant it is, whether the answer follows the STAR method if the answer has confident words and if the answer has filler words or technical terms. For interviews the system also checks if the answer has the right technical ideas and role-specific knowledge. The system keeps the results of each interview so that users can see how users performance changes over attempts. After an interview ends the system gives feedback about the candidate's performance. The system points out strengths and weaknesses highlights areas that need work and offers ideas for making answers better. The system can also give a version of an answer so that users see how a response could be presented more clearly. This module therefore helps users grow knowledge, communication skills, confidence and overall interview readiness through regular practice and personal feedback.

COMMUNICATION EVALUATION MODULE

The communication evaluation module looks at what you write or say and checks communication traits. It gives scores for grammar, clarity, vocabulary, confidence and relevance then produces a communication score. Mistakes

that appear are. A polished version of your answer is shown, helping you learn how to express yourself more clearly.

CODING AND APTITUDE MODULE.

In the Coding Practice component students can work on a range of programming problems. These problems are grouped by topic and difficulty level making it easier to progress step by step. Students can choose their preferred programming language—Java, Python or JavaScript—. Write their code directly in the platform. They can run their solutions. See how they perform against test cases. The system checks each submission using test data and provides feedback on execution status, score, number of test cases passed, total test cases and execution time. This helps students understand where they stand and what needs improvement.

The Aptitude Assessment component gives students timed tests focused on areas like quantitative aptitude, logical reasoning and verbal ability. Once the test is completed the system evaluates all answers automatically. It tracks how many questions were answered correctly incorrectly and left blank. It also records the time taken to finish the test. Based on this data performance across categories is analyzed. This allows students to see which topics they are strong in and which ones need attention.

AI ENGINE

The AI engine is a part of the system and works in two ways. The default mode uses a built-in rule-based engine that does not need an API or internet. You can also set up a large language model for smarter AI work. If the external AI fails the system falls back to the rule engine. The AI is used for resume analysis, interview review and skill suggestions.

PERSONALIZED LEARNING MODULE

The personalized learning module takes the data from all your practice to find the skills you need to improve. It keeps a snapshot of scores from each module. Uses those numbers to give focused suggestions. The learning path lists topics, resources, practice questions, completion status, priority and difficulty level. This lets the learning plan change based on how you perform by giving every user the same material.

DATA MODULE

The Performance Module gathers the results from parts of the system and saves them using Supabase. The dashboard pulls this data to show the user's placement-readiness score, skill performance, progress, personalized learning paths and recent activities. Records of past resume analysis, interviews, coding practice, aptitude assessments and

communication evaluations are kept in the database. This allows for tracking of performance, over time.

The method brings all placement preparation activities together in one system and links assessment with improvement. Than judging resume, interview, coding, aptitude and communication, on their own the system stores all those results and combines them to spot skill gaps and give tailored advice. This united approach lets students steadily raise their placement readiness.

ADMIN MODULE

The Admin Module manages and monitors the overall platform. It enables the administrator to manage the user accounts, uploaded resumes, job descriptions, coding problems and aptitude questions. The administrator can add, edit, delete and update the questions based on topic and difficulty level. Coding questions can be categorized based on programming languages, topics and difficulty levels, while aptitude questions can be categorized into quantitative aptitude, logical reasoning and verbal ability. The administrator can also enter the correct answers, test cases, explanations, and time limits for questions. Besides, the administrator can track the user performance, interview results, coding assessments, aptitude scores and communication evaluations saved in the database. This module helps to maintain the quality and accuracy of the questions and ensures students receive updated practice content.

IV. RESULT AND ANALYSIS

The AI-Powered Interview Preparation and Resume Coach was created to offer a platform for students who are getting ready for job placements. The system brings together different tools that help students prepare. These tools include checking resumes scoring them for how they match job requirements matching resumes to job descriptions practicing interviews evaluating communication skills practicing coding, testing aptitude, looking at skills gaps and offering personalized learning. The data collected from these tools is. Studied to give a full picture of how ready a student is for a job. The resume tool takes details from a resume and gives scores checks for keywords, evaluates how well the resume will pass through an ATS system and offers tips on

how to improve. The job matching tool compares a resume to a job description and shows which skills and keywords match and which are missing. The interview tool looks at how a student answers questions and gives scores, feedback and shows what the student is good at and what needs improvement. The communication tool checks grammar, clarity, word choice, confidence and how relevant the answers are. The coding and aptitude tools track how well a student does in tests how accurate their answers are, how test cases they pass and how long it takes them. These results help find out what skills a student is missing and create learning plans that're right for them. The dashboard shows

how ready a student is for a job how well they are doing in areas their progress each week what learning path they are, on and what they have been doing recently. This allows students to keep track of their preparation all the time. So the analysis shows that this system offers an personalized way to prepare for job placements by linking assessments tracking performance and making improvements in one place.

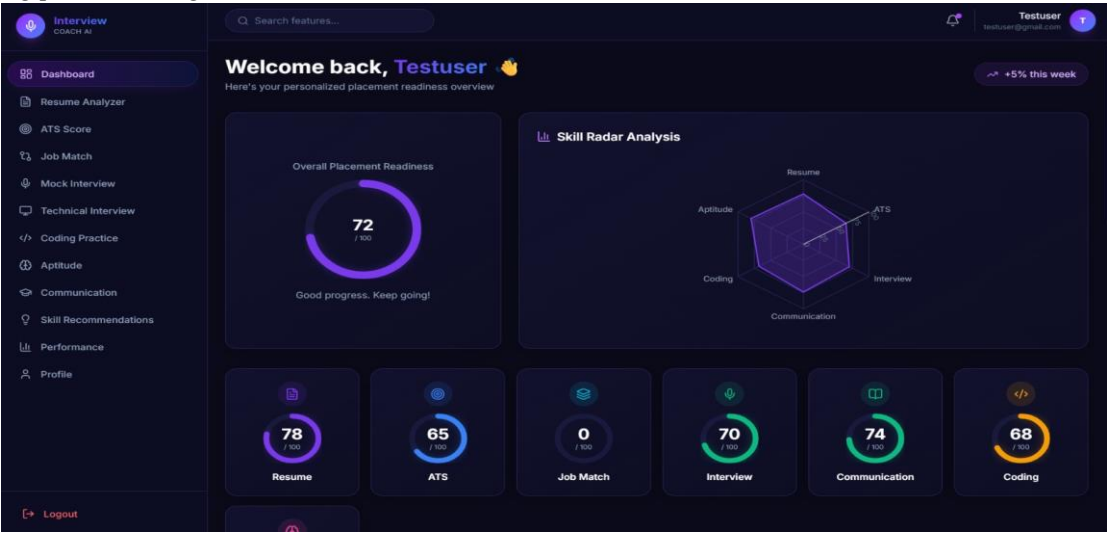


Fig. 4.1. Personalized placement readiness dashboard displaying overall readiness score and skill-wise performance analysis.

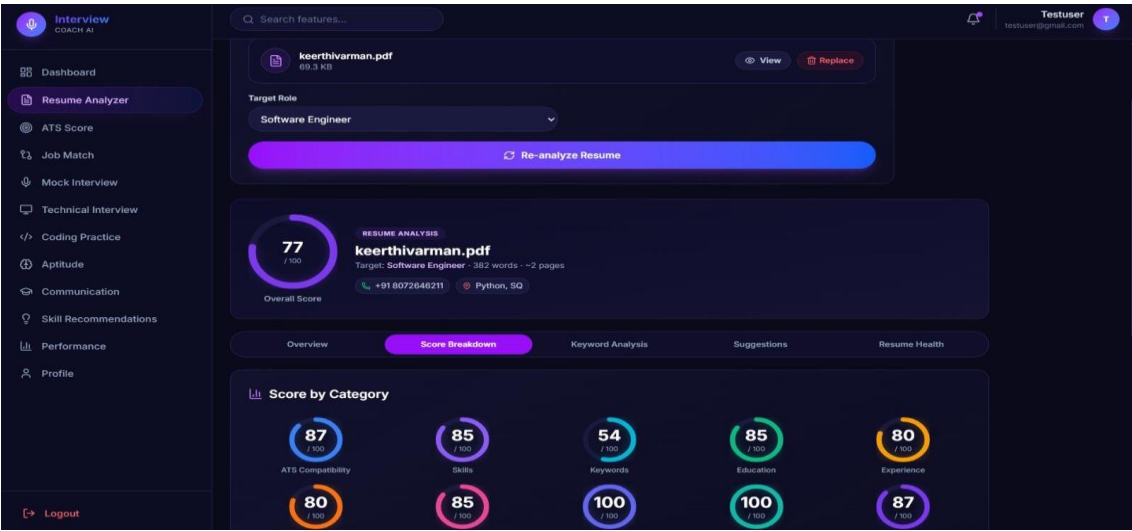


Fig. 4.2. Resume analysis results showing the overall resume score and category-wise score breakdown.

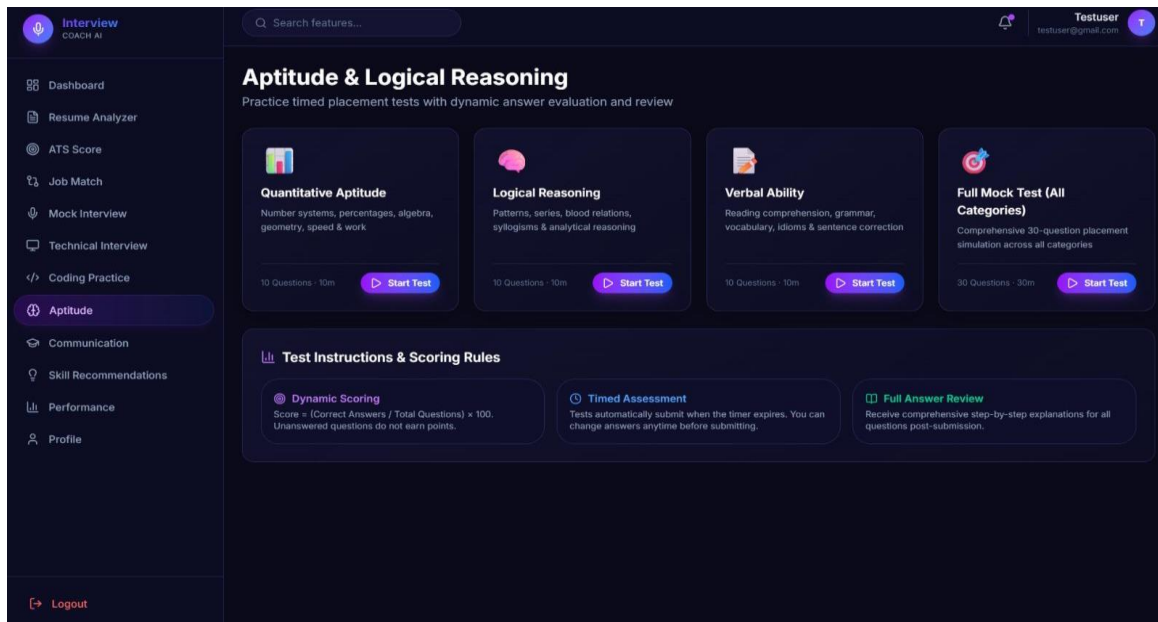


Fig. 4.3. Aptitude and logical reasoning assessment module.

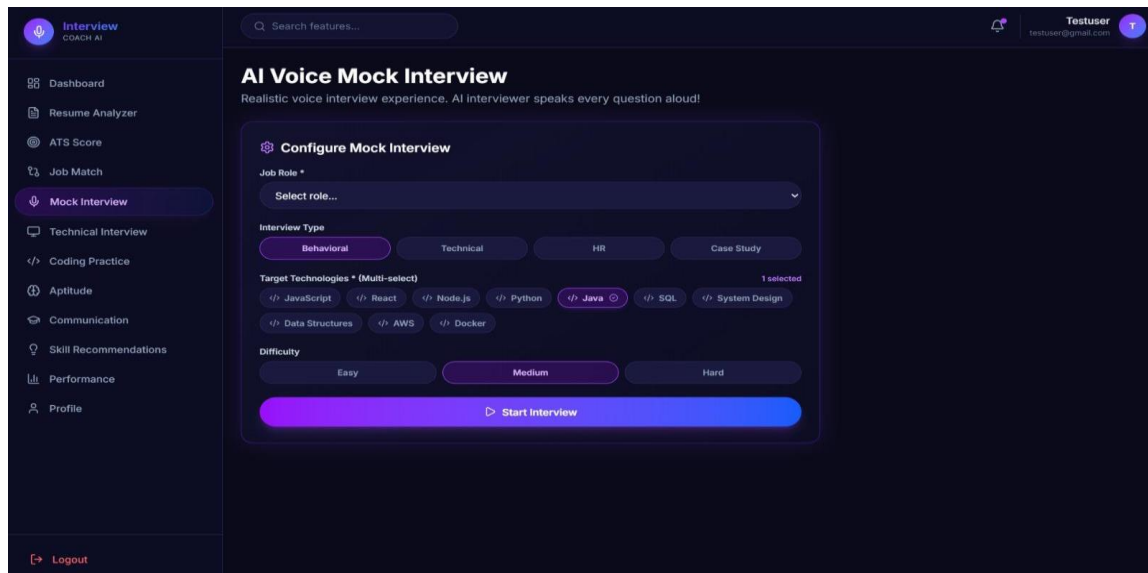


Fig. 4.4. AI voice mock interview configuration interface.

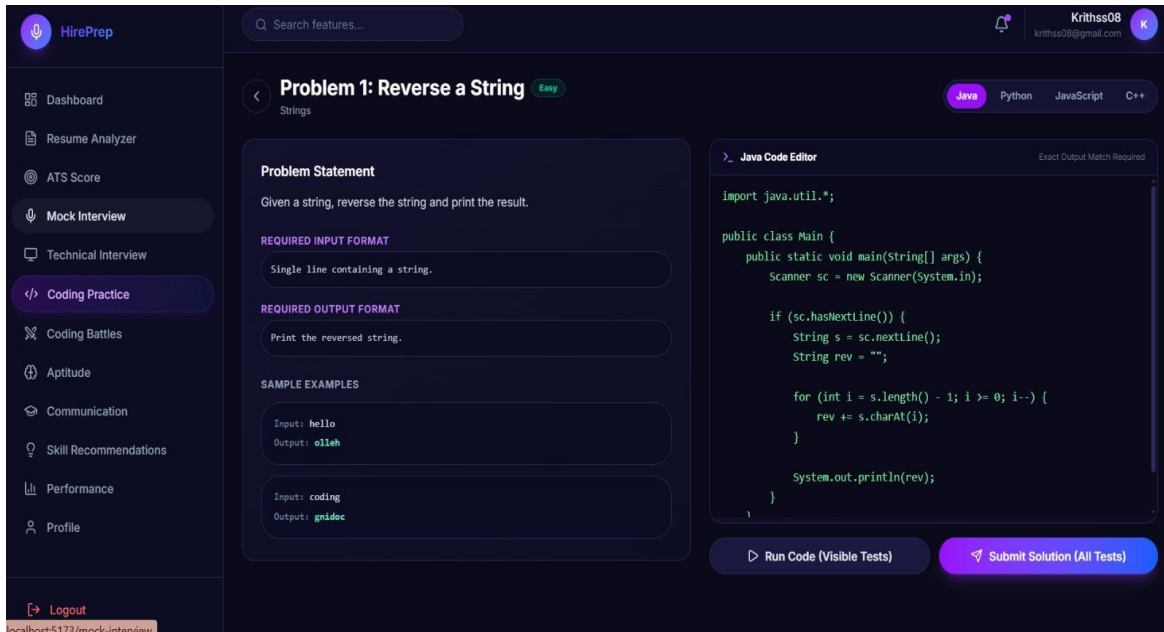


Fig. 4.5. Coding practice interface with problem statement and code editor.

V. CONCLUSION

The proposed AI-Powered Interview Preparation and Resume Coach is a platform designed to help students get ready for every part of the technical placement process. It brings together tools in one place, including resume analysis, ATS scoring, job description matching, mock interviews, technical interview practice, communication evaluation, coding exercises, aptitude tests, skill-gap detection and customized learning paths. All of these work to give students clear feedback and advice based on their performance. Using a dual-mode AI system with both a rule-based engine and optional large language model support gives the platform flexibility. This allows it to handle types of tasks—like analyzing resumes or evaluating interview responses—with consistent accuracy and adaptability. The system stores all user data in Supabase PostgreSQL so students can easily track their progress over time. They can see which areas need work and focus on improving them. The platform offers recommendations and learning plans tailored to each student's strengths and weaknesses. This helps users improve faster by focusing on what matters most to them. To keep user information safe the system includes security features such as authentication, password hashing, rate limiting, file validation and ownership checks. These protect data from unauthorized access or misuse. Overall the system provides a smart and data-

driven way to prepare for placements. It helps students build confidence and grow skills step by step leading to results, in real interviews and hiring processes.

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